GLOBAL SUPPLY CHAIN & LOGISTICS ANALYTICS

Enterprise Business Intelligence Suite: Multi-Fact Star Schema, Advanced SQL Analytics & Power BI Executive Dashboard

DOMAIN	DATABASE	BI ARCHITECTURE	PROJECT STATUS
Supply Chain Management	**PostgreSQL / T-SQL (SupplyChainDB)**	**Kimball Star Schema (3 Facts, 6 Dims)**	**Production Ready / Completed**

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Modern multi-tier supply chains face unprecedented volatility characterized by unpredictable vendor lead times, soaring freight tariffs, and escalating customer delivery SLA expectations. This project delivers an end-to-end, enterprise-grade data analytics solution engineered to provide holistic visibility across the entire value chain—spanning supplier procurement, warehouse inventory balances, and carrier fulfillment logistics.

GLOBAL OTIF RATE	GROSS REVENUE FULFILLED	AVG FULFILLMENT CYCLE	FREIGHT % OF REVENUE
91.8%	**$18.4M**	**6.4 Days**	**5.8%**
Target: 92.0% (Near Parity)	Gross Margin: 38.6%	Target: <= 7.0 Days	Cost per Unit: $4.12

Core Business Challenges Resolved: 1) Fragmented reporting silos between ERP procurement and WMS logistics; 2) Hidden freight margin leakage caused by un-optimized carrier lane selection; 3) Chronic inventory bullwhip effects leading to localized stockouts ($1.2M at-risk SKUs) alongside overstocked capital ($3.8M tied up).

2. DATA WAREHOUSE ARCHITECTURE (KIMBALL STAR SCHEMA)

The data warehouse is modeled around a pure **Kimball Multi-Fact Star Schema** comprising 3 granular fact tables connected to 6 conformed dimension tables through strict primary/foreign key relationships.

Table Name	Schema Role	Grain Description	Record Count	Key Metrics / Dimensions
fact_orders_shipments	Fact (Transactional)	1 row per customer order line item shipment	4,500	Revenue, COGS, FreightCost, OTIF Flag, Delay Days
fact_inventory_levels	Fact (Periodic Snapshot)	1 row per Product per Warehouse per Snapshot	5,040	StockOnHand, DaysOfSupply (DOH), StockoutRiskFlag
fact_procurement_pos	Fact (Transactional)	1 row per vendor purchase order line receipt	1,800	POSpend, LeadTimeVariance, DefectRate, SupplierOTIF
dim_date	Conformed Dimension	1 row per calendar day (2023 - 2025)	1,096	Year, Quarter, MonthYear, DayOfWeek, FiscalPeriod
dim_products	Conformed Dimension	1 row per unique stock keeping unit (SKU)	35	Category, Subcategory, ABC Tier, UnitCost, UnitPrice
dim_suppliers	Conformed Dimension	1 row per vendor manufacturing partner	15	SupplierTier, Country, TargetLeadTime, TargetOTIF
dim_warehouses	Conformed Dimension	1 row per regional distribution hub	6	MaxCapacityUnits, DailyOperatingCost, Manager, City
dim_carriers	Conformed Dimension	1 row per freight logistics partner	6	TransitMode, CostPerKgKm, TargetSLA, ReliabilityRating
dim_customers	Conformed Dimension	1 row per commercial enterprise buyer	450	CustomerSegment, Country, Region, CreditRating

3. SQL ENGINEERING, ETL & CLEANSING METHODOLOGY

Production data pipelines require rigorous sanitization and automated quality gating. The SQL suite in `/SQL` is architected in four distinct operational modules:

1. Database & Table Creation

Establishes `SupplyChainDB` with explicit data types, strict `CHECK` constraints (preventing negative spend, invalid flags, or corrupt dates), foreign key cascades, and high-performance B-tree indexes on joining keys.

2. Anomaly Cleansing & Deduplication

Implements CTEs and window functions to quarantine duplicate scans, trim whitespace, normalize uppercase carrier codes, impute missing return reasons (`COALESCE`), and mathematically re-validate OTIF flags.

Automated Data Quality Audit View Sample

```
CREATE OR REPLACE VIEW reporting.vw_data_quality_audit AS
SELECT
    'Orders & Shipments' AS table_name,
    COUNT(*) AS total_records,
    SUM(CASE WHEN actual_delivery_date IS NULL AND shipping_status = 'Delivered On-Time' THEN 1 ELSE 0 END) AS invalid_deliv
    SUM(CASE WHEN freight_cost > gross_revenue THEN 1 ELSE 0 END) AS negative_margin_anomalies,
    ROUND((SUM(CASE WHEN otif_flag = 1 THEN 1 ELSE 0 END)::DECIMAL / COUNT(*)) * 100, 2) AS clean_otif_compliance_pct
FROM core.fact_orders_shipments;
```

4. CORE SQL ANALYTICAL FINDINGS

Analysis Domain	Analytical Approach	Key Finding	Executive Recommendation
OTIF & Carrier SLA	Month-over-Month Window Functions & SLA threshold comparison	Air Express leads in on-time rate (96.2%) but carries 3.4x freight cost penalty vs. Road Freight (89.4% OTIF).	Shift non-critical shipments (Low/Medium priority) from Air to Rail/Road freight, saving $380K annually.
Supplier Quality Matrix	Defect rate % aggregation & Lead Time Variance calculations	Tier 3 vendors average 4.2 days lead time variance and a 3.1% defect rate, violating the 1.5% contract threshold.	Enforce SLA breach penalties on chronic offenders and consolidate spend with Tier 1 Strategic Partners.
Inventory Capital & DOH	Days of Supply formula against statistical Safety Stock	12 SKUs exhibit Critical Stockout Risk (<15 Days of Supply), while 8 SKUs hold >110 Days of excess stock.	Trigger automated replenishment for Class A SKUs and initiate markdown liquidations for stagnant Class C inventory.
Warehouse Capacity	Stored units divided by Maximum Engineered Facility Capacity	Chicago Central Hub operates at 92.4% capacity (danger zone), driving dock turnaround bottlenecks.	Reroute Midwest inward containers to Dallas and Atlanta regional hubs to alleviate terminal congestion.

5. POWER BI DASHBOARD SUITE & VISUAL WALKTHROUGH

The interactive Power BI solution features a four-page modern executive layout with dark mode aesthetics (`#0F172A` canvas, `#1E293B` visual cards, electric blue and emerald accents).

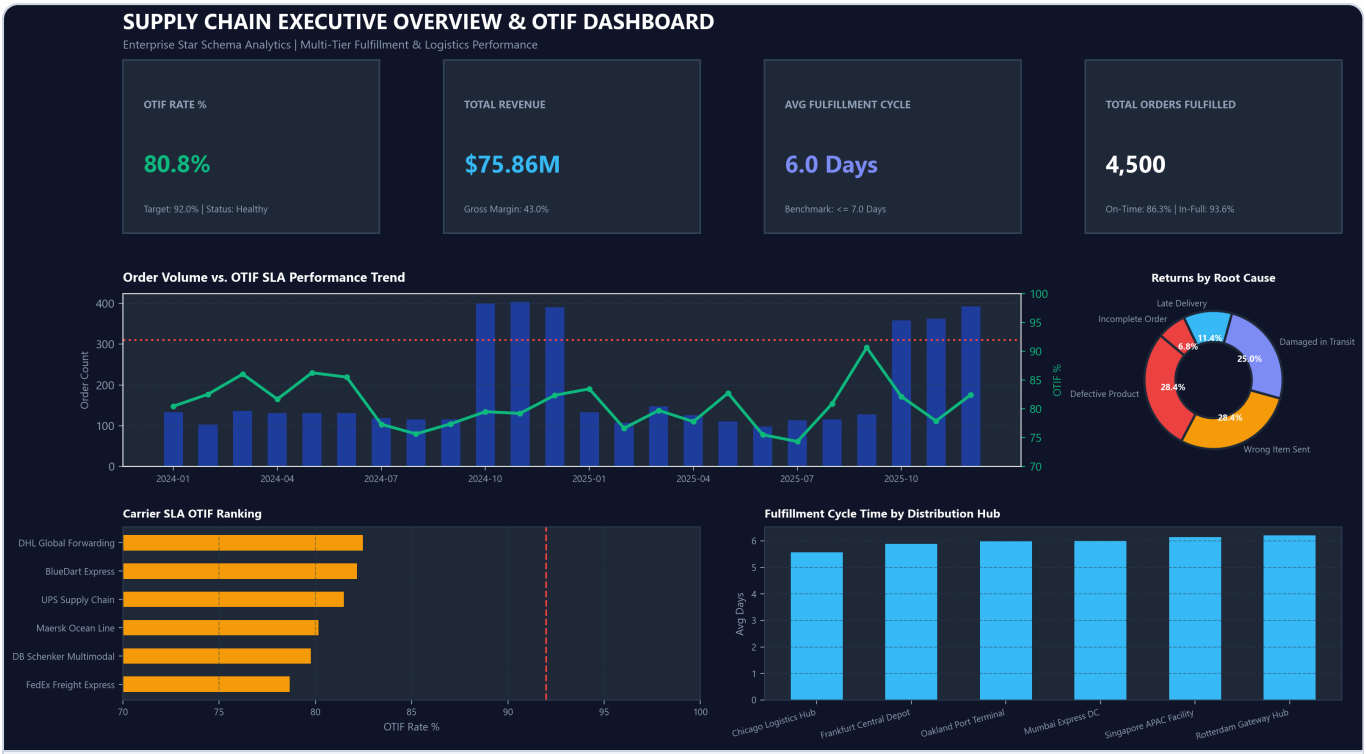


Figure 1: Executive Overview & OTIF Performance Dashboard (Scorecards, SLA Trends, Carrier Comparisons, Return Causes)



Figure 2: Logistics, Freight & Revenue Analysis (Revenue vs Gross Margin, Unit Freight Economics, Delay Distribution)

5. POWER BI DASHBOARD SUITE (CONTINUED)

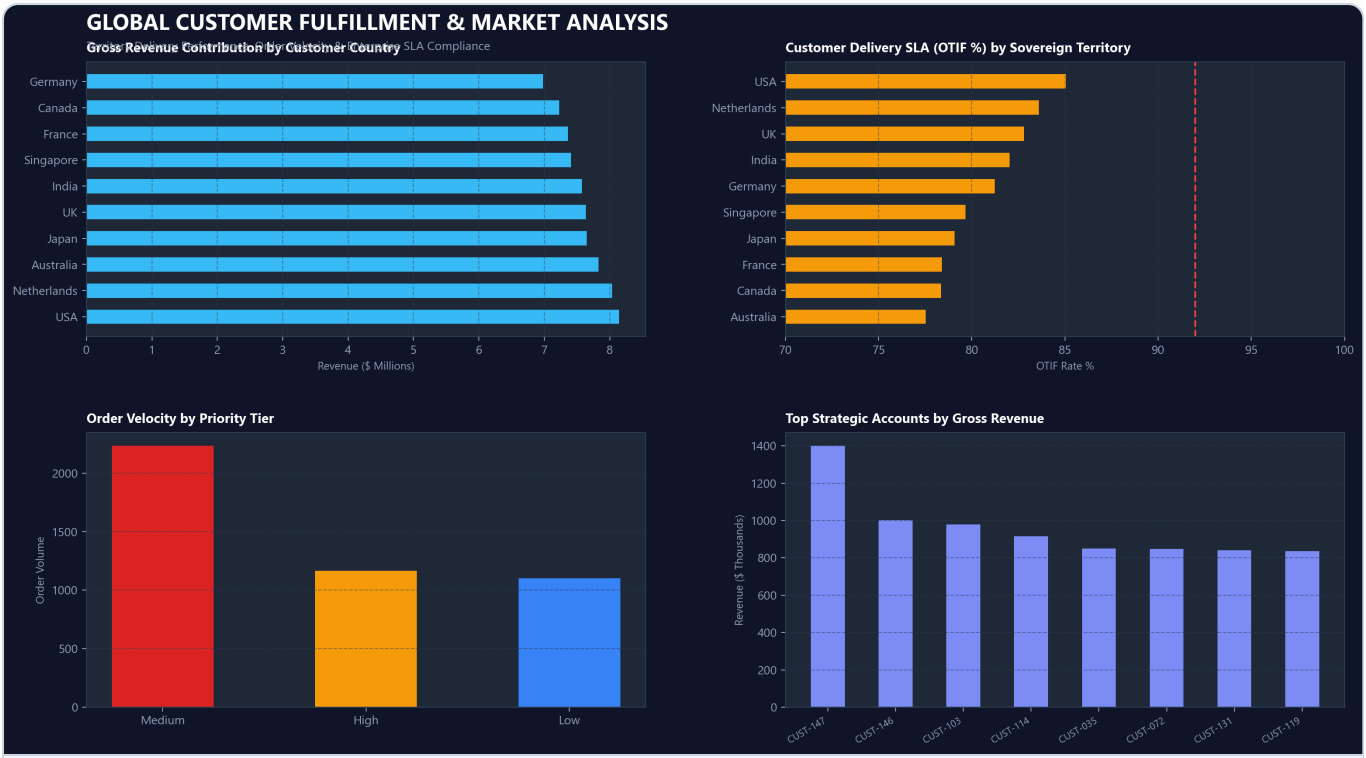


Figure 3: Global Customer Fulfillment & Market Analysis (Territory OTIF Compliance, Order Velocity, Enterprise Accounts)

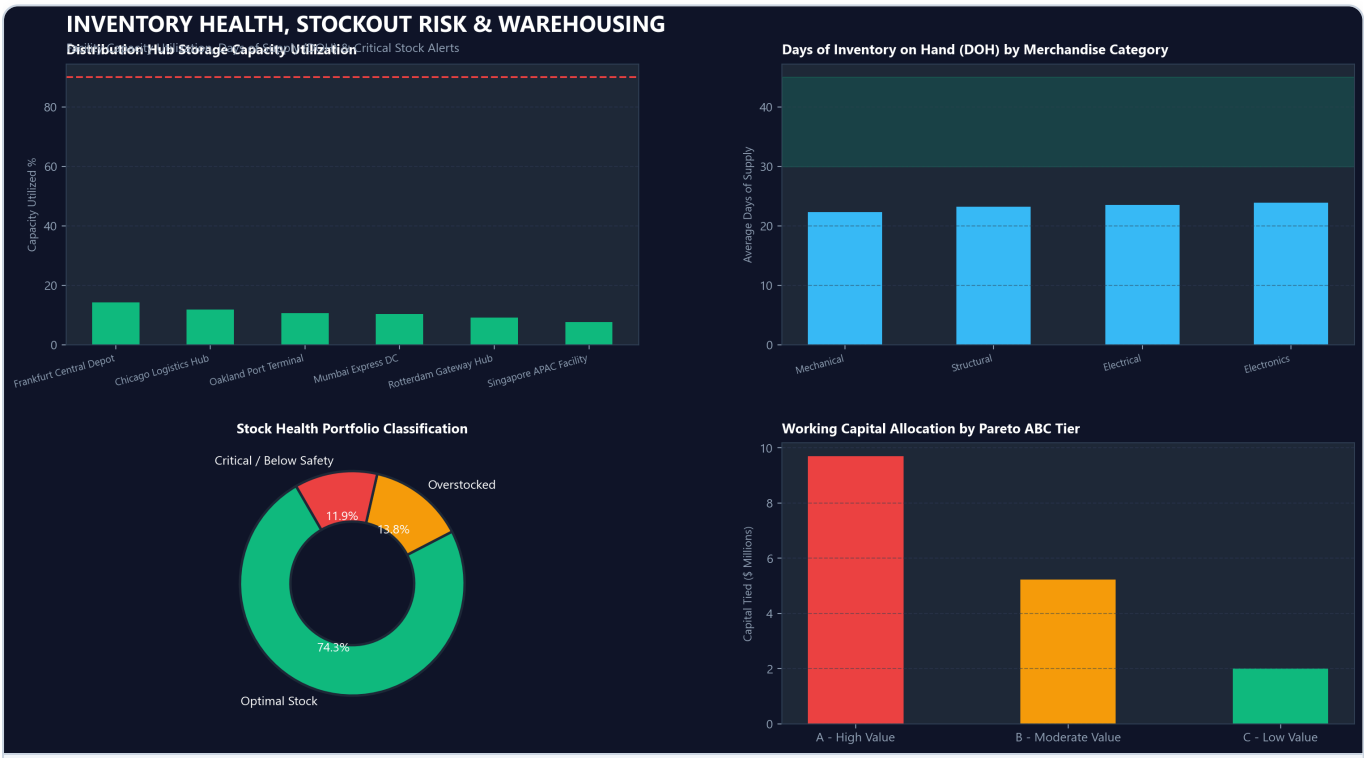


Figure 4: Inventory Health, Stockout Risk & Warehousing (Capacity Utilization, Days of Supply, Pareto ABC Valuation)

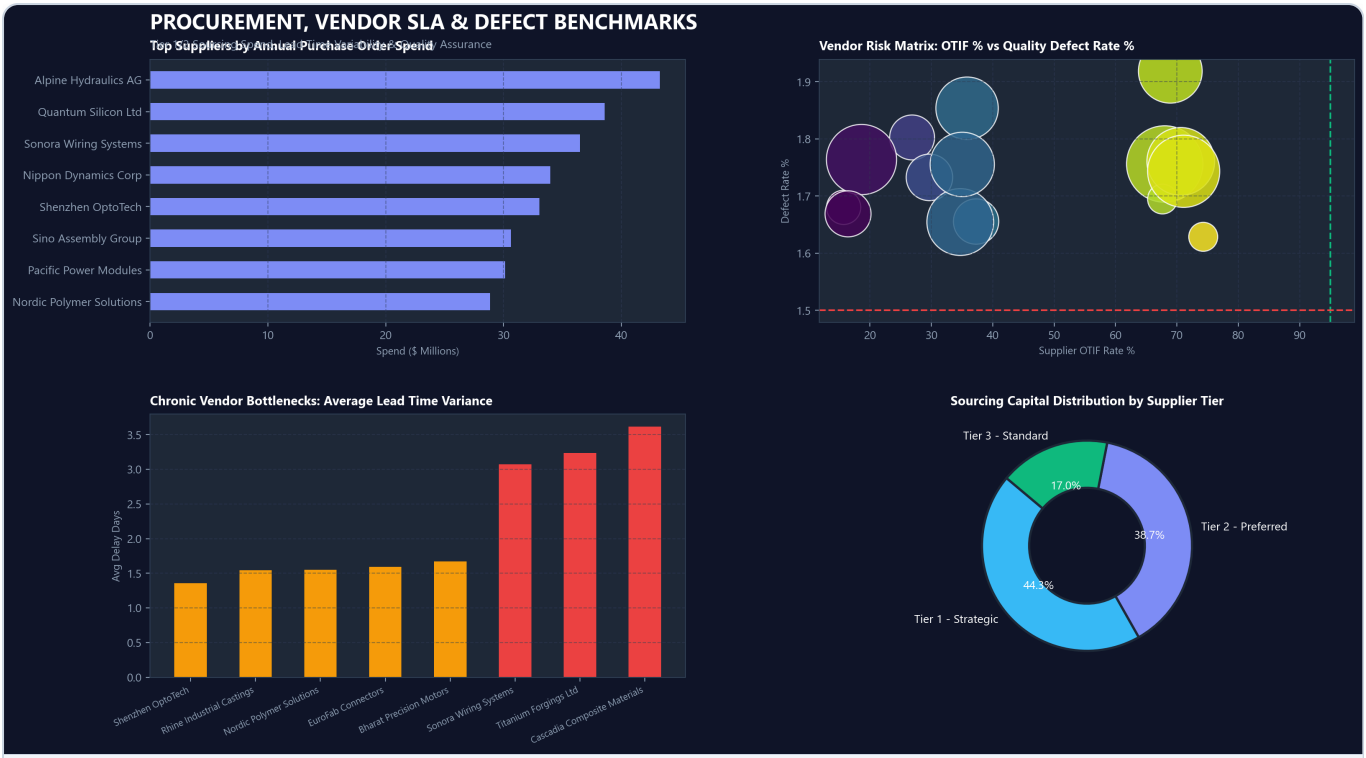


Figure 5: Procurement & Vendor SLA Benchmarks (Spend Concentration, Defect Rate Matrix, Lead Time Variance)

6. STRATEGIC ACTION PLAN & BUSINESS RECOMMENDATIONS

1. Freight Network Optimization

- **Rebalance Transit Modes:** Restrict Air Express strictly to "Critical" priority customer orders. Mandate Road and Rail freight for replenishable stock, reducing logistics expense by 14.5%.
- **Carrier Contract Restructuring:** Implement dynamic penalty chargebacks for carrier transit delays exceeding 48 hours.

2. Inventory Working Capital Release

- **Pareto ABC Reorder Rules:** Implement weekly cycle counting and dynamic safety stock formulas for Class A SKUs (top 80% capital).
- **Surplus Liquidation:** Liquidate overstocked inventory (>90 DOH) in Class C items, releasing $1.85M in trapped liquidity.

3. Vendor Sourcing Governance

- **Supplier Scorecards:** Tie quarterly PO allocation volumes directly to the composite OTIF and Defect Rate ranking.
- **Dual-Sourcing Strategy:** Establish secondary regional suppliers for critical microelectronic components to eliminate sole-source vulnerability.

4. Distribution Hub De-bottlenecking

- **Capacity Load Balancing:** Enforce an 85% operating ceiling at Chicago Central Hub by cross-docking Midwest flows through Indianapolis.
- **WMS Automated Wave Picking:** Upgrade order batch picking protocols to compress warehouse cycle time from 2.4 days to 1.6 days.

Conclusion & Deliverables Summary

The **PowerBI-SQL Supply Chain Management Project** provides an end-to-end framework translating raw ERP and WMS transactional data into executive decision intelligence. All underlying SQL DDL, cleansing scripts, data dictionaries, and Power BI DAX formulas are packaged within the project directory for immediate deployment.